

Thus, we see that the thrust required for steady level flight is directly proportional to the gross weight of the aircraft. Furthermore, we see that the thrust required also depends on the ratio of lift to drag and on the thrust angle. The lift-to-drag ratio, commonly referred to as *L over D*, is a very important measure of an airplane's aerodynamic efficiency, and this ratio is a strong function of airspeed. Since the weight is clearly independent of airspeed, for any given value of α_T , the airspeed that results in the greatest *L/D* will also result in the lowest thrust required. To see how the thrust required varies with airspeed, we must examine how the ratio *L/D* varies with airspeed.

The total drag force can be written as the parasitic drag plus the induced drag,

$$D = \frac{1}{2} \rho V^2 S_w \left(C_{Dp} + \frac{\kappa_{D\Omega} (C_{L,\alpha} \Omega)^2}{\pi R_A} - \frac{\kappa_{DL} C_{L,\alpha} \Omega}{\pi R_A} C_L + \frac{1}{\pi e_s R_A} C_L^2 \right) \quad (3.2.5)$$

where V is the airspeed, S_w is the area of the main wing, e_s is the span efficiency factor, and R_A is the aspect ratio. The parasitic drag coefficient, C_{Dp} , is very nearly a parabolic function of angle of attack. Since lift coefficient is a linear function of angle of attack, the parasitic drag coefficient can be written as a parabolic function of C_L ,

$$C_{Dp} = C_{Dp0} + C_{Dp,L} C_L + C_{Dp,L^2} C_L^2 \quad (3.2.6)$$

Using Eq. (3.2.6) in Eq. (3.2.5) and combining terms from the parasitic drag with terms from the induced drag, the total drag can be written as

$$D = \frac{1}{2} \rho V^2 S_w \left(C_{D0} + C_{D0,L} C_L + \frac{C_L^2}{\pi e R_A} \right) \quad (3.2.7)$$

where e is called the *Oswald efficiency factor*. The Oswald efficiency factor for the entire aircraft is considerably less than the span efficiency factor for the wing, because it accounts for part of the parasitic drag in addition to the induced drag. From Eq. (3.2.7) we can write

$$\frac{L}{D} = \frac{\frac{1}{2} \rho V^2 S_w C_L}{\frac{1}{2} \rho V^2 S_w C_D} = \frac{C_L}{C_D} = \frac{C_L}{C_{D0} + C_{D0,L} C_L + \frac{C_L^2}{\pi e R_A}} \quad (3.2.8)$$

The aerodynamic coefficients in Eq. (3.2.8) can all depend on Mach number. However, at low Mach numbers this dependence is slight.

Even at low Mach numbers, the lift coefficient for steady level flight is a function of airspeed. For level unaccelerated flight at any given airspeed, the aircraft must be trimmed to fly at an angle of attack that will produce the correct amount of lift to just satisfy Eq. (3.2.2). If the angle of attack is too large, too much lift is generated at the given airspeed and the airplane will climb. If the angle of attack is too small, not enough lift is generated and the airplane will begin to sink.

Using the definition of lift coefficient with Eq. (3.2.2), we have